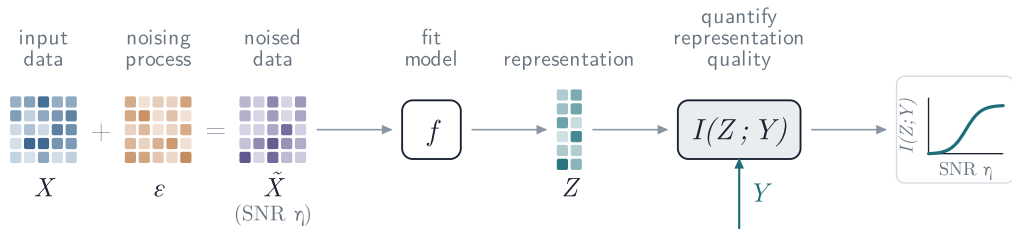


aauxiliary signal Y 
Spatial neighbors
MERFISH

Clonal neighbors
LARRY

Surface protein
CITE-seq

Developmental time
time-series scSeq
b

Data modality	Tag	Auxiliary signal	N samples	Mean UMI / cell	Model families	Noising process
Single-cell & spatial transcriptomics	Spatial	Spatial neighbor expression (MERFISH)	60,000	367	scVI Geneformer STATE PCA Random projection	Transcript count downsampling
	Clonal	Clonal neighbor expression (LARRY)	100,000	2,580		
	Protein	Protein abundance (CITE-seq)	100,000	8,100		
	Temporal	Embryo stage	10,000,000	2,500		
Images	TissueMNIST	Kidney cell type (fluorescence microscopy)	236,386	—	MobileNetV2	Pixel-wise additive Gaussian; pixelation
	Caltech101	Image label	2,707	—		
Protein sequences	Spike	Collection month (SARS-CoV-2 spike)	63,374	—	ESM2	Amino-acid substitution error